

Psychological Well-Being and Job Satisfaction as Predictors of Job Performance

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The happy-productive worker hypothesis has most often been examined in organizational research by correlating job satisfaction to performance. Recent research has expanded this to include measures of psychological well-being. However, to date, no field research has provided a comparative test of the relative contribution of job satisfaction and psychological well-being as predictors of employee performance. The authors report 2 field studies that, taken together, provide an opportunity to simultaneously examine the relative contribution of psychological well-being and job satisfaction to job performance. In Study 1, psychological well-being, but not job satisfaction, was predictive of job performance for 47 human services workers. These findings were replicated in Study 2 for 37 juvenile probation officers. These findings are discussed in terms of research on the happy-productive worker hypothesis.

If you want to be happy, be. —Alexei Konstantinovich Tolstoi, *Collected Works of Kosma Prutkov* (1884)

One of the most persistent topics of human interest appears to be the pursuit of well-being or “happiness” (B. Russell, 1930). Generally speaking, when researchers use the term *happiness*, they are usually referring to an individual’s psychological or subjective well-being (Diener, 1984; Diener, Suh, Lucas, & Smith, 1999). Although not identical constructs, these terms are often used interchangeably, but scholars prefer the term *well-being* to avoid the imprecision and lay connotation captured in the looser term *happiness* (Diener, 1984). Psychological well-being has three defining characteristics. First, well-being is a phenomenological event (Diener, 1994; Parnucci, 1995). In other words, people are happy when they subjectively believe themselves to be so. Second, well-being involves some emotional conditions. In particular, psychologically well people are more prone to experience positive emotions and less prone to experience negative emotions (Argyle, 1987; Diener & Larsen, 1993; Larsen & Diener, 1992; Warr, 1987, 1990). Third, well-being refers to one’s life as a whole. It is a global evaluation (Diener, 1994; Myers, 1992; Veenhoven, 1988).

The happy-productive worker hypothesis has

intrigued organizational scholars at least since the seminal Hawthorne experiments (Roethlisberger & Dickson, 1939). According to this hypothesis, “happy” workers demonstrate higher levels of job-related performance behaviors than do “unhappy” employees (Brief, 1998; Spector, 1997). Generally speaking, this hypothesis has typically been operationalized by correlating employee self-ratings of job satisfaction with supervisory ratings of performance. Thus, for many applied researchers, job satisfaction and happiness have become closely linked together (Wright & Doherty, 1998).

Over the years, several authors have provided comprehensive reviews of the satisfaction-performance relationship (see Brayfield & Crockett, 1955; Iaffaldano & Muchinsky, 1985; Locke, 1976; Petty, McGee, & Cavender, 1984; Vroom, 1964). For instance, Vroom (1964) observed in his classic study that the median correlation between these two variables was a modest .14. Unfortunately, interpretation of this result is subject to speculation, as Vroom inappropriately mixed both employee-level and work-group-level studies in his reported results. As a consequence, different conclusions are obtained when one removes the group-based studies from Vroom’s analysis.¹ A later, highly influential meta-analytic review by Iaffaldano and Muchinsky (1985) reported an average true score correlation of .17 between job satisfaction and job performance. However, this correlation represents the average of the correlations between each of the facet measures of satisfaction

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¹ We gratefully acknowledge the insightful comments of an anonymous reviewer regarding the interpretation of these earlier meta-analyses.

(i.e., pay, coworker, promotion, etc.) and performance. As a consequence, when one confines the analyses to a measure of overall job satisfaction, the corrected correlation is actually .29 (Judge, Thoresen, Bono, & Patton, 1998). Additional findings supportive of a relationship appeared in another meta-analysis by Petty et al. (1984), who found an average corrected correlation of .31 between a measure of overall job satisfaction and performance. Unfortunately, Organ (1988, chapter 4) cautioned that Petty et al.'s analysis might have included performance criteria that were contaminated by indexes of citizenship behavior.

Though the size of the satisfaction–performance relationship is still contested, many organizational researchers have seemingly relegated the notion of a happy–productive worker to the folklore of management (Judge, Hanisch, & Drankowski, 1995; Staw, 1986). We propose that this relegation may be premature and suggest the possibility that these ambiguous findings are a result of the different ways in which happiness has been operationalized in organizational research. To address this limitation of previous research, we report the results of two field studies that provide an opportunity to simultaneously examine job satisfaction and psychological well-being as predictors of performance.

Happiness and Job Satisfaction

As noted above, the most common means of operationalizing the happiness component of the happy–productive worker hypothesis in organizational research have been through the measurement of job satisfaction. In fact, speculation that job satisfaction predicts performance dates back to the early days of industrial/organizational psychology (Kornhauser & Sharp, 1932). Although there may be nothing inherently wrong with equating a happy worker with a satisfied worker, in so doing, one is required to make two tacit assumptions. First, because job satisfaction is specific to one's job, it does not include aspects of one's life outside of work. This relatively narrow scope stands in contrast to research on psychological well-being in which the happiness component is typically considered as a broader construct than job satisfaction, one that refers to aspects of an individual's life as a whole (Diener, 1984).

The second assumption involves the manner in which job satisfaction has been typically measured in organizational research. Although job satisfaction has been operationalized in many different ways, it usually is considered to be an attitude (Weiss &

Cropanzano, 1996). Moreover, in any investigation of job-related attitudes, Weiss and Cropanzano noted that it is important to separate the belief, or cognitive, component from the emotional, or affective, component. More specifically, this suggests that job satisfaction is based partially on what one feels and partially on what one thinks. Thus, according to Brief (1998), job satisfaction can best be defined as "an internal state that is expressed by affectively and/or cognitively evaluating an experienced job with some degree of favor or disfavor" (p. 86). However, the most widely used job satisfaction measures (i.e., the Minnesota Satisfaction Questionnaire, the Job Descriptive Index, etc.) contain few, if any, affectively toned scale items (Brief & Roberson, 1989). Alternatively, happiness, as the term is commonly understood, is primarily an affective or emotional experience. In other words, happy individuals *feel* good in the sense that they experience a good deal of positive emotion and, relatedly, less negative emotion (Warr, 1987, 1990).

We propose that a continued emphasis on the job satisfaction–performance relationship does not constitute an adequate test of the happy–productive worker hypothesis. More specifically, whereas job satisfaction is a relatively narrow construct that is usually operationalized cognitively, happiness generally refers to a broader construct that has primarily affective overtones. As a consequence, a number of researchers are now concerned with the possible role of psychological well-being in the prediction of performance. To that end, we note that our examination of job satisfaction and psychological well-being as tests of the happy–productive worker thesis should only be considered as descriptive in nature. More specifically, as stated earlier, we are not proposing that either job satisfaction or psychological well-being is equivalent to happiness, only that they are two widely used organizationally based operationalizations of worker happiness. We now develop the theoretical basis for why a relationship may exist between psychological well-being and performance.

Psychological Well-Being and Performance

Psychological well-being is usually defined in terms of the overall effectiveness of an individual's psychological functioning (Gechman & Weiner, 1975; Jamal & Mitchell, 1980; Martin, 1984; Sekaran, 1985). More specifically, on the basis of the circumplex model of emotion, psychological well-being measures the hedonic or pleasantness dimension of individual feelings (J. A. Russell, 1980; J. A.

Russell, Weiss, & Mendelsohn, 1989). Clinical psychologists have long recognized the role of the pleasantness dimension of well-being (i.e., happiness vs. sadness or depression) in the determination of various individual outcomes. For example, depressed individuals have very low self-esteem, tend to be pessimistic, and exhibit reduced motivation and slowed thought processes (Holmes, 1991; Wright & Bonett, 1997). Furthermore, unlike job satisfaction, which is centered around the work context, psychological well-being is a broader construct. Most typically, psychological well-being is considered as a primarily affective-based "context-free" or global construct. Unlike various measures of job satisfaction, well-being is not tied to any particular situation (Kornhauser, 1965; Warr, 1987, 1990).

Like their clinical psychology and public health counterparts, organizational theorists have also long recognized the extensive costs, in both human and financial terms, attributable to employee dysfunctional psychological well-being (George, 1992; Quick, Quick, Nelson, & Hurrell, 1997). For instance, depression, loss of self-esteem, hypertension, alcoholism, and drug consumption have all been shown to be related to work-related dysfunctional psychological well-being (Ivancevich & Matteson, 1980). Because these variables have, in turn, been related to declines in work outcomes (Quick et al., 1997), it is possible that psychological well-being and employee performance are related (Wright, Bonett, & Sweeney, 1993).

More direct organizational evidence that psychological well-being is related to employee performance comes from two studies by Wright and his colleagues. Wright et al. (1993) found that psychological well-being was positively related to supervisory ratings of performance. In a second study, Wright and Bonett (1997) also found a significant relationship between employee well-being and job performance. Furthermore, previous research has shown a significant relationship between well-being and job satisfaction (e.g., Diener et al., 1999; Judge & Locke, 1993). However, to date, no published empirical research has provided a comparative test of job satisfaction and psychological well-being as predictors of performance. Using two independent field samples, we conducted the present research to attempt to fill this void in the literature. First, in a pilot study, we examined the relationship of psychological well-being and job satisfaction as predictors of performance. On the basis of certain methodological limitations of Study 1, we replicated and extended our findings in Study 2. Taken together, these two studies

provide an initial opportunity to simultaneously examine the relative contribution of psychological well-being and job satisfaction as predictors of job performance.

Study 1: Well-Being and Composite Job Satisfaction as Predictors of Job Performance

The purpose of Study 1 was to directly assess the relationships of well-being and job satisfaction to job performance. In this cross-sectional field study, participants completed measures of psychological well-being and a composite measure of job satisfaction. Supervisory ratings of employee performance were obtained. In view of previous work, we made the following three predictions:

- Hypothesis 1: Composite job satisfaction will be positively related to supervisory ratings of job performance.
- Hypothesis 2: Psychological well-being will be positively related to job performance ratings.
- Hypothesis 3: The relative contributions of psychological well-being and composite job satisfaction as predictors of performance are examined.

Method

Participants

Through a direct-contact procedure, human services workers employed by a northern California county agency ($N = 47$) were asked to participate in Study 1. The actual sample included all 47 employees, representing a response rate of 100%. These college-educated, human services workers provided face-to-face direct care, service, and supervision to clients assigned to them. All respondents were employed within the same department, performed the same job duties, were mostly male (74%), had a mean age of 39 years, and had a mean job tenure of 10.8 years.

Measures

Psychological well-being. As a measure of psychological well-being, this study used the eight-item Index of Psychological Well-Being developed by Berkman (1971b). The Berkman scale uses many of the same items as Bradburn and Caplovitz's (1965) earlier measure, but with a more general time horizon. Respondents were asked how often they felt "very lonely or remote from other people," "depressed or very unhappy," "bored," "so restless you couldn't sit long in a chair," "vaguely uneasy about something without knowing why," "particularly excited or interested in something," "pleased about having accomplished something," and "on top of the world." The reader is referred to Berkman (1971a, 1971b) and Wright and Bonett (1992) for a more complete description of the scoring and validation of the index. In this study, the coefficient alpha was .72.

Job performance. Job performance was measured through the use of supervisory ratings. Because all of the employees in the study ultimately reported to the top-ranking administrative officer in the department, this departmental manager rated each employee's performance. The measurement device was a one-item, overall performance rating scale that asked the question: "Overall, how would you rate this employee's performance over the past year?" Employees were rated using a 5-point scale that ranged from 1 (*poor*) to 5 (*excellent*).

Job satisfaction. Satisfaction was measured by three widely recognized dimensions: degree of satisfaction with the work itself, degree of satisfaction with coworkers, and degree of satisfaction with supervision (Price & Mueller, 1986). The satisfaction items asked the following questions: "All in all, how satisfied are you with the work itself of your job?" "All in all, how satisfied are you with your coworkers?" and "All in all, how satisfied are you with the supervision?" Ratings were made on a 5-point scale ranging from 1 (*very unsatisfied*) to 5 (*very satisfied*). Theoretical and empirical justification for combining these three facet measures of satisfaction into a composite measure have been widely documented (Price & Mueller, 1986; Wright & Bonett, 1991). The coefficient alpha obtained in Study 1 was .63.

Results and Summary Discussion

Correlational Analyses

Table 1 displays means, standard deviations, and the intercorrelations for the Study 1 variables. Table 1 also provides the data necessary to test Hypotheses 1 and 2. Hypothesis 1 predicted that composite job satisfaction would be positively related to supervisory evaluations of performance. This prediction was not supported. Hypothesis 2 predicted a positive relationship between psychological well-being and performance. In support of Hypothesis 2, a significant positive correlation was obtained between well-being and supervisory evaluations of job performance ($r = .32, p < .05$).

Regression Analyses

For our present purposes, Hypothesis 3 is the most important. Hypothesis 3 proposed that we examine

the relative contribution of composite job satisfaction and psychological well-being as predictors of job performance. In testing Hypothesis 3, we had to control for some possible confounding variables. For example, even though all of the employees were employed within the same department and performed the same job duties, they did vary in terms of age, gender, and tenure. Cranny, Smith, and Stone (1992) noted the potential relationships between job satisfaction and such demographic variables as age, gender, and tenure. Brush, Mock, and Pooyan (1987) conducted a meta-analysis of 19 studies and found a mean correlation between age and job satisfaction of .22. In addition, Spector (1997) noted that gender may be an important factor to consider in conducting research on job satisfaction. Berkman (1971b) also noted that similar relationships may exist among these demographic variables and well-being. Finally, the possibility of relationships among age, gender, tenure, and performance has also been noted (Blum & Naylor, 1968). To address these issues, we performed two separate hierarchical regression analyses (Cohen & Cohen, 1983). The first assessed the contribution of well-being above and beyond that of job satisfaction and the control variables. The second assessed the contribution of job satisfaction above and beyond that of well-being and the control variables.

Incremental Contribution of Well-Being

We assessed the incremental contribution of well-being in two steps. In Step 1, job performance was regressed on age, gender, job tenure, and composite job satisfaction. As shown in Table 2, these four variables failed to account for a significant amount of the variance in job performance, $F(4, 41) = 0.60, ns$. In Step 2, well-being was added to the equation, $F(5, 40) = 2.14, ns$. The change in R^2 from Step 1 to Step 2 represents the increase in job performance variance explained by well-being above and beyond what was accounted for by the control

Table 1
Means, Standard Deviations, and Intercorrelations for Study 1 Variables

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5	6
1. Age	39.0	7.0	—	.73**	-.17	-.12	.08	-.01
2. Job tenure	11.0	5.3		—	-.00	-.24	-.08	.13
3. Gender ^a					—	.14	.13	-.08
4. Composite job satisfaction	3.3	0.7				—	.35**	-.08
5. Psychological well-being	3.6	1.5					—	.32*
6. Job performance	4.1	0.7						—

^a Gender was dummy coded 1 for *male* and 2 for *female*.

* $p < .05$. ** $p < .01$. All tests two-tailed.

Table 2
Hierarchical Regression Assessing the Incremental Contribution of Well-Being, Study 1

Independent variable	Step 1		Step 2	
	β	SE	β	SE
Intercept	4.88**	.83	6.42**	.95
Age	-.02	.02	-.04*	.02
Gender	-.14	.23	-.25	.22
Job tenure	.04	.03	.05*	.03
Composite job satisfaction	-.05	.15	-.16	.14
Employee well-being			.19**	.07
	$R^2 = .06$		$R^2 = .21$	
	Adjusted $R^2 = .00$		$\Delta R^2 = .15^{**}$	
			Adjusted $R^2 = .11$	

* $p < .05$, one-tailed. ** $p < .01$, one-tailed.

variables. As shown in Table 2, the change in R^2 was significant, resulting in a change in $R^2 = .15$, $t(40) = 2.81$, $p < .01$, indicating that well-being did account for a significant proportion of the variance in job performance above and beyond that accounted for by the control variables.

Incremental Contribution of Job Satisfaction

In the second hierarchical regression analysis, the control variables (age, gender, job tenure, and employee well-being) were entered at Step 1, $F(4, 41) = 2.35$, *ns*. The composite job satisfaction dimensions were added during Step 2, $F(5, 40) = 2.14$, *ns*. As displayed in Table 3, composite job satisfaction did not account for a significant amount of variance in job performance, resulting in a nonsignificant change in $R^2 = .03$, $t(40) = 1.12$, *ns*, above and beyond the effect of the control variables.

Summary

Support was obtained for Hypothesis 2. In addition, these results are consistent with the notion that well-being is a predictor of job performance, even when controlling for job satisfaction, age, gender, and tenure (Hypothesis 3). Unfortunately, Study 1 also contains certain methodological limitations as described below. These prompted us to conduct a second investigation.

Study 2: Well-Being, Composite Job Satisfaction, and Composite Performance

A concern in Study 1 involved the reliance on a single-item measure of performance as our dependent variable. A second problem was the marginal alpha (.63) obtained for our composite measure of job satisfaction. To further investigate these methodologi-

Table 3
Hierarchical Regression Assessing the Incremental Contribution of Job Satisfaction, Study 1

Independent variable	Step 1		Step 2	
	β	SE	β	SE
Intercept	5.84**	.80	6.42**	.95
Age	-.04*	.02	-.04*	.02
Gender	-.28	.22	-.25	.22
Job tenure	.06*	.03	.05*	.03
Employee well-being	.17**	.07	.19**	.07
Composite job satisfaction			-.16	.14
	$R^2 = .18$		$R^2 = .21$	
	Adjusted $R^2 = .11$		$\Delta R^2 = .03$	
			Adjusted $R^2 = .11$	

* $p < .05$, one-tailed. ** $p < .01$, one-tailed.

cal issues, we conducted a second field study. Participants again completed measures of psychological well-being and composite job satisfaction, whereas four job-related measures of performance were completed for each employee by their supervisor. More specifically, using the evaluative procedure validated by Wright and his colleagues (Wright & Bonett, 1993; Wright et al., 1993), management personnel from the current organization confirmed four dimensions as appropriate and relevant for assessing employee performance: work facilitation, goal emphasis, team building, and support. Past research and the results of Study 1 allowed us to again formulate the following three predictions:

Hypothesis 1: Composite job satisfaction will be positively related to supervisory ratings of composite job performance.

Hypothesis 2: An individual's feelings of well-being will be positively related to supervisory ratings of composite job performance.

Hypothesis 3: The relative contribution of psychological well-being and composite job satisfaction as predictors of a composite measure of performance are examined.

Method

Participants

Juvenile probation officers from a West Coast county agency ($N = 37$) were asked to participate in Study 2 through a direct contact procedure. The actual sample included 37 employees, representing a response rate of 100%. All of the respondents were college graduates, were employed within the same department, and performed the same job duties. The mean age was 39.6 years. The sample was 78% male and 22% female.

Measures

Psychological well-being. Well-being was measured using the same Berkman (1971a, 1971b) Index of Psychological Well-Being scale that was used in Study 1. Study 2's coefficient alpha was .70.

Job satisfaction. Three widely recognized dimensions of job satisfaction (degree of satisfaction with the work itself, degree of satisfaction with coworkers, and degree of satisfaction with supervision) were again used to measure job satisfaction (Price & Mueller, 1986). Ratings were made on a 5-point scale ranging from 1 (*very unsatisfied*) to 5 (*very satisfied*). As noted, the theoretical and empirical justification for combining these three facet measures of satisfaction have been widely documented (Price & Mueller, 1986; Wright & Bonett, 1991). The coefficient alpha obtained in Study 2 was .72.

Job performance. Management personnel from the current organization confirmed four dimensions as appropriate and relevant for assessing employee performance: support, work facilitation, goal emphasis, and team building. These four dimensions were measured using a 5-point scale ranging from 1 (*never*) to 5 (*always*) regarding the extent to which the employee emphasized a particular dimension. All employees ultimately reported to, and are evaluated by, the same top-ranking administrative officer. This departmental officer provided ratings of each employee's work performance for the relevant 1-year review period. Following Wright and Bonett (1993), the four items, each examining a separate performance dimension, were summed to form a composite measure of performance (coefficient $\alpha = .87$).

Results

Correlational Analyses

The means, standard deviations, and intercorrelations for each of the Study 2 measures are reported in Table 4. Hypothesis 1 predicted a relationship between composite job satisfaction and performance. This prediction was not supported. Hypothesis 2 predicted a positive relationship between psychological well-being and performance. In support of Hypothesis 2, a significant positive relationship was obtained between psychological well-being and composite performance ($r = .34, p < .05$).

Regression Analyses: Hypothesis 3

As in Study 1, Hypothesis 3 is the most important for our present purposes. Hypothesis 3 proposed that we examine the relative contribution of job satisfac-

Table 4
Means, Standard Deviations, and Intercorrelations for Study 2 Variables

Variable	<i>M</i>	<i>SD</i>	1	2	3	4	5
1. Age	39.6	7.0	—	-.11	-.04	.21	.02
2. Gender ^a				—	.18	.22	-.00
3. Composite job satisfaction	3.3	0.9			—	.10	.08
4. Psychological well-being	3.6	1.4				—	.34*
5. Composite performance	3.5	0.8					—

^a Gender was dummy coded 1 for *male* and 2 for *female*.

* $p < .05$. All tests two-tailed.

Table 5
Hierarchical Regression Assessing the Incremental Contribution of Well-Being, Study 2

Independent variable	Step 1		Step 2	
	β	SE	β	SE
Intercept	3.16**	1.00	4.53**	1.15
Age	.00	.02	-.01	.02
Gender	-.03	.32	-.18	.32
Composite job satisfaction	.07	.15	.05	.15
Psychological well-being			.20*	.09
	$R^2 = .01$		$R^2 = .13$	
	Adjusted $R^2 = .00$		$\Delta R^2 = .12^*$	
			Adjusted $R^2 = .02$	

* $p < .05$, one-tailed. ** $p < .01$, one-tailed.

tion and psychological well-being as predictors of job performance. We were again able to control for various demographic variables in Study 2, in this case for employee age and gender. Two separate hierarchical regression analyses were performed to address Hypothesis 3 (Cohen & Cohen, 1983). The first assessed the contribution of psychological well-being above and beyond that of composite job satisfaction and the control variables. The second assessed the contribution of composite job satisfaction above and beyond that of psychological well-being and the control variables.

In the first analysis, composite performance was regressed on well-being, composite job satisfaction, age, and gender. More specifically, at the first step in the analysis, the control variables of composite job satisfaction, age, and gender were entered into the equation simultaneously. As shown in Table 5, these variables failed to account for a significant amount of variance in composite performance, $F(3, 33) = 0.08$, *ns*. At the second step, all of the remaining variables (well-being and control) were entered simulta-

neously, $F(4, 32) = 1.20$, *ns*. The change in R^2 from Step 1 to Step 2 represents the increase in composite performance variance explained by the study variable, well-being, above and beyond what was accounted for by the control variables. As shown in Table 5, the change in R^2 was .12, $t(32) = 2.13$, $p < .05$, indicating that well-being did account for a significant proportion of the variance in composite performance above and beyond that accounted for by the control variables.

In the second hierarchical regression analysis, composite performance was regressed on well-being, job satisfaction, age, and gender. More specifically, at the first step in the analysis, the control variables of psychological well-being, age, and gender were entered into the equation simultaneously. As shown in Table 6, considered together, these variables failed to account for a significant amount of variance in composite performance, $F(3, 33) = 1.61$, *ns*. However, we note that a significant relationship was obtained between psychological well-being and composite performance, $t(33) = 2.20$, $p < .05$. At the

Table 6
Hierarchical Regression Assessing the Incremental Contribution of Job Satisfaction, Study 2

Independent variable	Step 1		Step 2	
	β	SE	β	SE
Intercept	4.69**	1.03	4.53**	1.15
Age	-.01	.02	-.01	.02
Gender	-.17	.31	-.18	.32
Composite job satisfaction			.05	.15
Psychological well-being	.20*	.09	.20*	.09
	$R^2 = .13$		$R^2 = .13$	
	Adjusted $R^2 = .05$		$\Delta R^2 = .00$	
			Adjusted $R^2 = .02$	

* $p < .05$, one-tailed. ** $p < .01$, one-tailed.

second step, all of the remaining variables (composite job satisfaction and controls) were entered simultaneously, $F(4, 32) = 1.20, ns$. As shown in Table 6, the change in R^2 was .00, $t(32) = 0.33, ns$, indicating that composite job satisfaction did not account for a significant proportion of the variance in composite performance above and beyond that accounted for by the control variables. Taken together, these two analyses clearly demonstrate the relative contribution of psychological well-being, but not composite job satisfaction, in the prediction of composite performance in the present research.

General Discussion

The two independent studies reported here provided consistent results: Psychological well-being was related to job performance ratings. These findings lend support to the basic proposition that "happy" workers often have higher performance but that "happiness" should be operationalized as psychological well-being. More specifically, in both Studies 1 and 2, we found that well-being was related to performance ratings beyond the effect of composite job satisfaction.

These results warrant further discussion. Psychological well-being was shown to be more predictive of performance than job satisfaction. We propose that one reason for this finding may be that job satisfaction provides less of a test of happiness, because job satisfaction scales typically do not contain any items that directly assess happiness. As a consequence, organizational researchers are increasingly realizing the possibility that job satisfaction is not an appropriate operationalization of happiness at work.

Given the modest sample sizes in the two studies, it would not be appropriate to accept the failure to find a *significant relationship between job satisfaction and performance* without additional research endeavors. For example, Judge et al. (1998) noted in their meta-analysis that the job satisfaction–performance correlation varies greatly by occupation and level of job complexity. More specifically, they found a stronger correlation in jobs higher in complexity and that allow for increased worker decision discretion and autonomy. Although both samples in our study involved college-educated employees, both samples also involved occupations (human services and probation officer) that are greatly constrained by numerous bureaucratic rules, regulations, and procedures. Thus, before one can make any definitive conclusions, we recommend that further research be conducted in which occupational differences in job

complexity and decision latitude can be systematically controlled.

The present research does provide additional support for a growing body of knowledge suggesting a key role of psychological well-being in the prediction of performance (Wright & Bonett, 1997; Wright et al., 1993). However, the present research designs were cross-sectional in nature. What is now needed is research that examines the predictive impact of change in employee well-being over time (Adler & Weiss, 1988). Long-term, prospective longitudinal research designs would allow for the period of time between psychological well-being and such important organizational outcome variables as performance and turnover to more naturally and developmentally unfold.

The present studies are not without their own methodological limitations. Perhaps the most important source of potential bias concerns our performance measures. It is possible that psychologically well people are simply more pleasant to be around. As such, they might receive performance ratings that are biased in a positive direction (e.g., Staw & Barsade, 1993). Our criteria involved supervisory measures of performance and might be subject to this type of bias. However, it should be emphasized that job satisfaction was not associated with job performance in either Study 1 or Study 2. If rating bias was accounting for the psychological well-being and job performance relationship, then one would also have expected significant bivariate relationships between job satisfaction and performance. These were not obtained. However, whenever possible, we recommend that future research include more objective or quantitatively oriented measures of performance.

Practically speaking, Wright and Cropanzano (1998) proposed that the call for more quantitatively based performance measures might prove quite a difficult task for future research endeavors, however. For instance, Wright et al. (1993) noted that many organizations emphasize non-task-specific performance dimensions in the performance appraisal process. As a case in point, in the present research, the appraisal processes used in our research samples are based solely on supervisory evaluations. In fact, in both of the present employee samples, employee groups have consistently and successfully lobbied to keep objective measures of performance out of the performance review process. Furthermore, anecdotal evidence obtained by one of the article authors indicates that this may be a rather prevalent occurrence in many organizations. This type of situation led Wright and Cropanzano (1998, p. 491) to

conclude that "regardless of whether supervisory performance evaluations include halo and other forms of bias, they are predictive of 'success' from the employees' point of view."

A second concern involves our measure of psychological well-being. Although the obtained reliability coefficients are certainly acceptable for research purposes (.72 and .70), they might be problematic in field situations in which the consequences involve the actual hiring or firing of employees based on their psychological well-being scores. In fact, in general, the improper use of various measures of psychological well-being could pose some difficult ethical issues for management personnel involved in selection, placement, training, and development decisions. For example, the failure to be selected because of one's psychological well-being could further depress or sadden the job candidate, which in turn could make the individual even less employable in the future. In sum, if psychological well-being is found to be predictive of performance across individuals and situations, any application of this finding to an organization's evaluation process will pose profound ethical and practical dilemmas for both the organizations and individuals involved.

A final potential methodological concern involves the manner in which satisfaction was measured in the present studies. Widespread acceptance exists for combining facet measures of satisfaction into composite measures (Price & Mueller, 1986; Wright & Cropanzano, 1998). However, Judge et al. (1998) found considerable variability in the correlations of different measures of job satisfaction with performance. That is, while Judge et al.'s results established that the mean corrected job satisfaction-performance correlation is quite close to .30 ($\pm .01$), the individual corrected correlations were much more variable ($.30 \pm .27$). Judge et al. proposed as one explanation for these results that there is no single job satisfaction-performance correlation. For instance, correlations ranged from $p = .06$ for the Faces scale to $p = .51$ for the Hoppock Job Satisfaction Blank. We recommend that future research be undertaken to examine why these measures are so differentially related to performance.

Despite these possible limitations and the need for additional research, it does seem that the generations of managers who believed that a happy worker is a productive worker may well have been correct, at least when happiness is operationalized as psychological well-being. That said, we close by emphasizing a large and important issue. This article was concerned with the relationships among job satisfaction, psycho-

logical well-being, and performance. However, this should in no way be taken to suggest that performance is the only, or even the most important, reason that well-being is important. In fact, too often in applied research, the emphasis appears to have been on what one might call the disease model. That is, a focus primarily concerned with fixing what is wrong with someone, as opposed to developing what is right. For example, Myers and Diener (1995) noted that psychological publications focusing on negative states outnumber their positive counterparts by a ratio of 17 to 1! However, it seems to us that promoting happiness or well-being is an intrinsic good for which all should work. This positive focus has been referred to in psychology as the health model (Ruack, 1999), or an approach which emphasizes the development of what is right with people. If this health approach promotes better performance, and our findings are consistent with a growing body of literature suggesting that this is so, then so much the better. Regardless, employee psychological well-being remains valuable for its own sake.

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